

Ema Maria Monteiro Martins

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Master's student in Informatics and Computing Engineering with a focus on **game development**, **virtual/augmented reality**, and **computer vision**. Experienced in **Unity** and **Godot**, building cooperative multiplayer games, VR/AR experiences, and gameplay systems, with additional experience applying **AI** to interactive applications.

Education

Universidade do Porto <i>Master's Degree in Informatics and Computing Engineering</i>	Sep 2024 - Jul 2026 Porto, Portugal
Universidade do Minho <i>Bachelor's Degree in Informatics Engineering (completed concurrently with Biomedical Engineering)</i>	Sep 2022 - Jul 2024 Braga, Portugal
Universidade do Minho <i>Bachelor's Degree in Biomedical Engineering (completed concurrently with Informatics Engineering)</i>	Sep 2020 - Jul 2024 Braga, Portugal

Experience

Mindera <i>Master Thesis Intern</i>	Feb 2026 - Jun 2026 Porto, Portugal
- Developed an internal project engineering an AI agent to act as a believable second player in two-player video games, prioritizing human-like behavior over flawless performance. - Integrated computer vision and AI models (NVIDIA NitroGen, YOLO, Grounding DINO) to process in-game environments in real-time.	

Projects

Village Retreat <i>Unity, Mixamo, FMOD</i>	Itch Link Repo Link
Co-designed the Game Design Document and developed a 3D cooperative exploration game where players complete four minigames to collect materials and build a house, ending with a final cutscene. Integrated FMOD audio and an AI-assisted asset pipeline (Meshy, Mixamo).	
Escape do Saber <i>Unity</i>	Repo Link
Developed an educational VR escape room for Meta Quest 2 with 3 mini-games (color mixing, anatomy, English comprehension) and an interactive graduation-cap ending.	
Balloon Popping Robot Competition <i>Unity, ML-Agents</i>	Repo Link
Engineered a MARL Battle Royale simulation in Unity, applying POCA and Curriculum Learning to train agents for obstacle avoidance and resource collection in a shrinking arena.	
Races <i>Unity, Vuforia AR</i>	Repo Link
Developed a mobile AR racing game with an AI opponent, motion-based controls, and a strategic phase for placing obstacles/power-ups via AR markers.	
Right Under Your Moustache <i>Godot</i>	Itch Link Repo Link
Co-developed a 2D puzzle-stealth game, programming special abilities, puzzle logic, and enemy detection, plus original visual assets and audio.	

Skills

Technical Skills	Languages
Game Dev: Godot, Unity	Portuguese: Native
AI & Machine Learning: Unity ML-Agents, NitroGen, YOLO, Grounding DINO	English: Fluent
Backend: C, C#, C++, Python, Java, Haskell, .NET	Spanish: Basic
Frontend: JavaScript, React, Vue.js, HTML/CSS, Pug	
Mobile: Kotlin, React Native, Flutter	
Databases: MongoDB, MySQL, Redis, PostgreSQL, Neo4j	
Practices: Git, Agile/Scrum	

Leadership & Volunteering

Volunteer – Extracurricular Classes <i>Volunteering: Taught extracurricular classes to children aged 6–9, promoting environmental awareness.</i>	Sep 2022 – Mar 2023 Braga, Portugal
Vice-president of Literatuna – Universidade do Minho <i>Student Organization Leadership: Coordinated activities and performances of a university musical group.</i>	Sep 2023 – Jul 2024 Braga, Portugal
Erasmus+ Project – ANIMA 2022 <i>International Sociocultural Event: International meeting of sociocultural animation.</i>	May 2022 Portugal / Europe